Sri Lanka Institute of Information Technology



Machine Learning | IT3091

TRIAGE-Telco: A Stability-Aware Machine Learning Framework for Reliable Customer Retention Decisions

-Final Group Submission-

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(Y3.S1. WE. DS. 0102)

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Declaration

Group ID: 2026-DS-07

This is the final report for Group 2026-DS-07's IT3091 Guided Data Track project, TRIAGE-Telco. It explains the business problem, dataset, data preparation, model experiments, prediction-disagreement analysis and evaluation of proposed customer-retention decisions.

The report acknowledges its data source, lecture materials and research references. The accompanying project package contains the notebooks and Python code, source dataset, data dictionary, decision logs and model comparisons. Section 13.1 provides the repository location, and Section 13.3 explains AI assistance with planning, debugging, review, writing and packaging.

The group is responsible for checking the work and interpreting its results accurately. We distinguish measured results from illustrative examples and future proposals, and openly report the limits caused by earlier test-data exposure and reuse of validation folds. Each member must submit an accurate individual Personal Learning Journey.

Abstract

TRIAGE-Telco studies customer churn prediction and model agreement for a telecom retention team with limited contact capacity. The dataset contains 7,043 records: 5,634 development customers and 1,409 historical-test customers. Fold-fitted preprocessing supports Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, CatBoost and an RBF Support Vector Machine (SVM). We compare class weights, random oversampling and SMOTENC, and examine feature choices, training–validation gaps, learning curves and nested SVM validation.

Tuned CatBoost achieved mean development average precision (AP) of 0.6727 and recorded historical-test AP of 0.6658. The later tuned SVM achieved nested outer-fold AP of 0.6555; SMOTENC reduced AP in all three tested classifier families. The recorded CatBoost, Gradient Boosting and tuned Random Forest panel therefore remains unchanged. Votes at probability 0.35 assign ACT, REVIEW, MONITOR or HOLD, while separate sensitivity tests and a Disagreement Atlas show where predictions vary.

At equal contact limits, TRIAGE did not reliably capture more historical churners than ranking by average model risk. We recommend risk ranking for contact priority and disagreement as review context. Stronger input checks and synthetic examples demonstrate a runnable scoring path, not proven retention benefit. Historical test exposure and reused model-selection folds limit validation claims. Nested validation assesses the SVM search only; future independent data and measured intervention outcomes remain necessary.

Acknowledgement

We thank the IT3091 Machine Learning teaching team at the Sri Lanka Institute of Information Technology for the lecture materials, assignment guidance and Guided Data Track telecommunications scenario. These resources provided the foundations for framing the retention problem and justifying the project's analytical choices.

We acknowledge the Telco Customer Churn dataset source, the authors of the cited research and the developers of the open-source software used for analysis, modelling and visualisation. Their resources supported the implementation and interpretation of the project. AI assistance is acknowledged separately in Section 13.3.

We thank the members of Group 2026-DS-07 for their contributions across business understanding and EDA, preprocessing and feature engineering, modelling and stability analysis, and evaluation and business interpretation.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
AP	Average precision
API	Application programming interface
AUC	Area under the curve
CB	CatBoost
CI	Confidence interval
CSV	Comma-separated values
CV	Cross-validation
DSL	Digital subscriber line
ECE	Expected calibration error
EDA	Exploratory data analysis
F1 / F2	F-scores balancing precision and recall; F2 weights recall more
FN / FP	False negatives / false positives
GB	Gradient Boosting
ID	Identifier
IQR	Interquartile range
L1 / L2	Absolute-value / squared-value regularisation penalties
LR	Logistic Regression
ML	Machine learning
OOF	Out-of-fold
PCA	Principal component analysis
PR-AUC	Precision–recall area under the curve; result files use this label for AP
RBF	Radial basis function
RF	Random Forest
ROC-AUC	Area under the receiver operating characteristic curve
ROI	Return on investment
ROS	Random oversampling
SD	Standard deviation
SMOTE	Synthetic Minority Over-sampling Technique
SMOTENC	SMOTE for mixed nominal and continuous features
SVM	Support Vector Machine
TN / TP	True negatives / true positives
TRIAGE	Telecom Retention Intervention and Agreement-Guided Evaluation

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1. Introduction

1.1 Background

Customer churn means a customer stops using a telecom provider's service. A retention team may want to identify customers at risk of leaving, but its time and budget for contacting people are limited. The Guided Data Track supplies the Telco Customer Churn dataset and asks us to make useful, evidence-based decisions from that information.

A conventional churn project might choose the one model with the best overall score. But two models with similar scores may warn differently about the same customer. TRIAGE-Telco studies these disagreements to see whether they provide useful context when deciding whom to contact or review.

1.2 Problem Statement

Predictions can change when we change the algorithm, its settings, random seed or way of handling the smaller churn class. These changes matter when a warning might lead to a customer contact. Importantly, disagreement does not prove an error, and unanimous agreement does not prove a correct prediction.

Research question: Can predicted churn risk and agreement between models help a telecom retention team make better decisions under a fixed contact limit? In particular, does TRIAGE find more actual churners than ordinary risk ranking at the same contact capacity?

1.3 Implemented Solution and Main Finding

We compared five classifier families in the original development workflow and added tuned SVM as a sixth family in a later development-only extension. Three retained models estimate churn probabilities; a voting rule converts these into ACT, REVIEW, MONITOR and HOLD. The added experiments test imbalance handling and generalisation without rewriting the historical panel or its recorded test results.

The contact-budget experiments did not show a reliable increase in identified churners compared with sorting customers by average model risk. Our recommendation is therefore to rank contacts by risk and use model disagreement to guide possible human review. The dataset does not tell us whether a contact would actually prevent churn.

1.4 Project Objectives

- Understand the dataset and identify quality issues and descriptive churn patterns.
- Compare defensible preprocessing choices and a baseline with alternative classifiers.
- Examine customer-level sensitivity to model and preprocessing choices.
- Evaluate prediction quality, review workload, subgroup errors and fixed-budget capture.
- Provide traceable code, decision evidence and a recommendation consistent with the results.

2. Assignment Track

Our group ends in 7, so we follow the Guided Data Track for Telecommunications. The assignment supplies the business scenario and dataset, while our group must choose the decision lenses, modelling task, preparation methods, validation, algorithms and final recommendation [1].

Each record describes a customer's contracts, services, billing and churn outcome. We use these fields to predict recorded churn, describe high-risk profiles and test whether model disagreement helps limited-capacity retention decisions.

3. Business Lenses

The assignment requires one primary business decision lens and allows one secondary lens when it directly supports the first. Our two lenses focus on the same retention decision.

3.1 Primary Lens - Stability-Aware Customer Retention Decision Support

Our primary lens asks which customers should receive retention attention and when staff should examine conflicting model predictions. In this report, 'stability' means that predictions remain consistent across the modelling choices we tested. It does not mean that a customer's behaviour was observed to be stable over time.

Two customers may have similar estimated churn risks but very different levels of model agreement. TRIAGE-Telco makes those differences visible. Because all models learn from related data, even a unanimous decision can be wrong.

Primary decision need: Which customers should receive retention attention under limited capacity, and how should disagreement inform human review?

3.2 Secondary Lens - Stable High-Risk Customer Profiling

The supporting lens profiles the customers placed in ACT, meaning all three selected models predict churn at the 0.35 voting threshold. It describes their contracts, tenure, services and charges so the retention team can understand who receives priority.

This directly supports the primary lens: it explains where the ACT rule concentrates attention. Many development ACT customers had month-to-month contracts and shorter tenure, but these observations do not prove that changing a contract will prevent churn.

The Disagreement Atlas supports these lenses by locating model disagreement in customer groups. It is a diagnostic analysis rather than a separate third business lens.

4. Machine Learning Task and Output

4.1 Machine Learning Task

Our main ML task is binary classification: estimate whether the recorded target is Churn = Yes (1) or Churn = No (0). Logistic Regression is the baseline. Decision Tree, Random Forest, Gradient Boosting and CatBoost form the original comparison; tuned SVM is an additional development challenger. We use saved folds and model-appropriate preprocessing.

We also compare predictions across model families, CatBoost settings, random seeds and imbalance treatments. These checks reveal how sensitive a customer's classification is to our choices; they are not a formal confidence interval for an individual prediction.

The models learn only the two-class churn task. ACT, REVIEW, MONITOR and HOLD are assigned afterward using a fixed rule based on the three selected models' votes, not learned as four training labels.

4.2 Input Variables

Feature Group	**Candidate Variables**
Customer Profile	gender, SeniorCitizen, Partner, Dependents
Customer Relationship	tenure, Contract
Phone / Internet Services	PhoneService, MultipleLines, InternetService
Security / Support / Add-ons	OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Billing and Payment	PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Identifier	customerID - identifier for joins and traceability only; excluded from predictors

There are 19 approved predictors. The unique customerID is kept only to connect records with predictions, never as an input to the model. Four additional engineered features were experimented with but were not adopted in the final reference input set.

4.3 Target Variable and Unit of Analysis

Each row represents one telecom customer. We encode Yes as 1 and No as 0. The dataset records churn status, but does not define a specific future time window such as 'will churn within 30 days'.

4.4 Data Quality and Potential Leakage

We identified 11 blank TotalCharges entries, all for customers with zero months of tenure. We preserved meaningful service categories and excluded customerID from predictors. Preprocessing parameters were learned from training folds, although earlier full-data EDA and historical test-data exposure limit independence.

4.5 Final Output - Retention Triage

Each of the three selected models estimates a churn probability for the same customer. A score of 0.35 or higher gives that model one positive vote. We keep all three scores, their mean, the vote count and the final ACT/REVIEW/MONITOR/HOLD state so staff can see both risk and agreement. Section 11 shows examples.

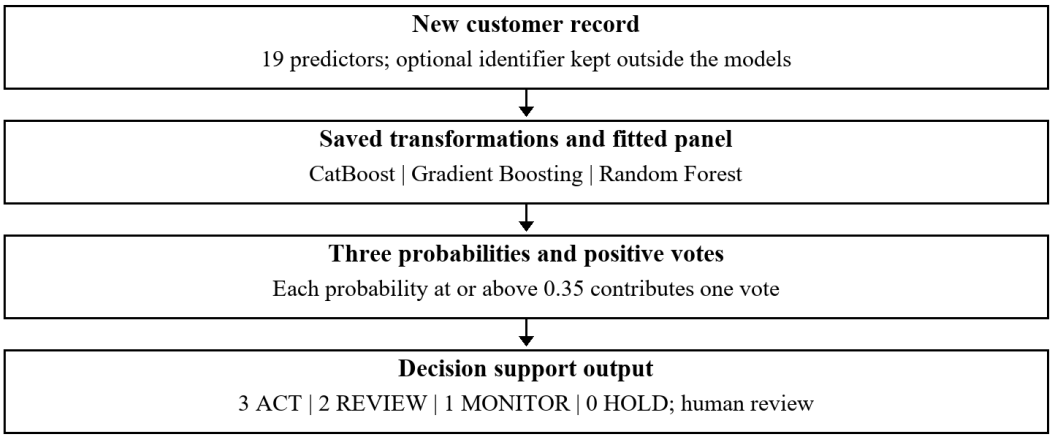


Diagram A. Implemented path from a new customer record to a reviewed decision.

Triage State	Implemented Vote Rule	Business Interpretation
ACT	3 of 3 positive votes	Prioritise for a reviewed retention contact.
REVIEW	2 of 3 positive votes	Review because the models disagree despite a positive majority.
MONITOR	1 of 3 positive votes	Monitor the split-vote case; the majority is negative.
HOLD	0 of 3 positive votes	Lower priority, with no guarantee that the customer will stay.

HOLD means low priority under the recorded policy—not a promise that the customer will stay. In the historical test, 96 actual churners were placed in HOLD. Agreement is useful context, not a guarantee of correctness.

5. Implemented Workflow

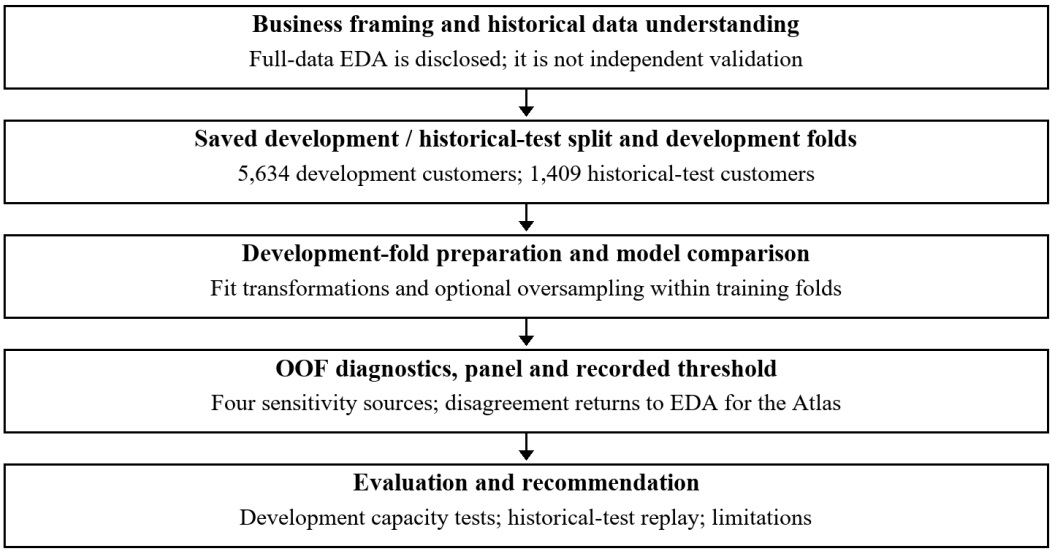


Diagram B. Development workflow, validation boundaries and feedback to EDA.

5.1 Problem Framing

- Stakeholder: telecom retention / customer-management team.
- Decision need: prioritise customers using predicted risk and agreement under limited contact capacity.
- Unit of analysis: one customer.
- Target output: churn risk + prediction stability + triage state.

Major choices are recorded with their alternatives, reasons and supporting evidence. The final decision-log summary and the detailed machine-readable logs are available in the project repository.

5.2 Data Understanding and Data Quality

A data dictionary explains what each field means. We checked whether records are complete, identifiers are unique, the target classes are balanced, and numeric values are plausible. Findings were logged before choosing data treatments.

5.3 Exploratory Data Analysis

Member 1 explored customer tenure, contracts, internet services, support options, payment methods and charges in all 7,043 historical records. The purpose was to discover patterns and identify questions for modelling, not to decide which customers must churn.

The early patterns are associations, not proof of why any person leaves. After modelling, the Disagreement Atlas revisited customer characteristics to find groups on which different models often disagree.

5.4 Data Preprocessing and Feature Engineering

Member 2 prepared inputs inside each training fold: numeric repairs, categorical encoding, scaling and imbalance experiments. Eleven blank TotalCharges values occur at zero tenure and receive the recorded zero treatment. Four engineered features were tested with Logistic Regression and, in the extension, Random Forest. Neither test supported adopting the complete engineered set as the default.

5.5 Baseline Model Development

We first fitted Logistic Regression as an understandable baseline. Its five-fold mean AP was 0.6617. More flexible models were then judged using the same development folds, so extra complexity had to be justified by results.

5.6 Alternative Model Development

Decision Tree, Random Forest, Gradient Boosting and CatBoost were the original alternatives. The completed extension adds an RBF SVM, which learns a margin-based decision boundary after scaling. These algorithms provide different modelling assumptions; the final probability-voting panel remains the original three selected tree ensembles.

5.7 Validation and Hyperparameter Exploration

The 5,634 development customers use five saved stratified folds. The original model searches reused those folds for selection, so their scores may be optimistic. A later nested SVM experiment uses three inner folds for tuning within each of the five outer training partitions. It evaluates the SVM search more independently, but does not remove earlier full-data exploration or historical-test exposure.

5.8 Prediction Stability and Reliability Analysis

We retained CatBoost, Gradient Boosting and tuned Random Forest because each was within an absolute 0.010 mean-fold AP of the best model. We also examined stricter (0.005) and wider (0.015) limits. This differed from the initial proposal's example 2% ROC-AUC rule, and we document the change. Close scores do not prove the models are statistically equivalent.

5.9 Model Evaluation

Member 4 checked how models rank customers (AP and ROC-AUC), which customers they misclassify (precision, recall, F1 and confusion counts), whether estimated probabilities match observed frequencies (calibration), and how well limited contact queues capture churners. The recorded test results were replayed without choosing a new policy from those outcomes.

5.10 Final Recommendation

Our practical recommendation is to contact people in order of estimated churn risk, while showing model disagreements to staff as an extra review signal. TRIAGE did not reliably capture more historical churners than ordinary risk ranking, and the available data cannot prove that outreach prevents churn.

6. Member 1 - Business Understanding and EDA

Jayasinghe S.S. | IT24102186

Jayasinghe's role was to understand what the dataset means for a telecom retention team. Member 1 checked data quality, described churn patterns and identified questions for later experiments. After model training, the Disagreement Atlas linked customer profiles with out-of-fold model disagreements. Personal contributions are addressed separately in each member's learning journey.

6.1 Business Framing and Role

Framing question	**Project answer**
Who uses the result?	A telecom retention or customer-management team with limited contact capacity.
What decision is supported?	Which customers should receive priority attention, and which predictions require human review because models disagree?
What is one observation?	One customer record. Churn is the Yes/No target; customerID is a join key rather than a predictor.
What counts as useful evidence?	Good churn ranking, understandable disagreement, reproducible comparisons and contact-budget performance.
What is outside the evidence?	Whether an offer actually prevents churn, how much profit it produces, and whether the system generalises to a new time period.

6.2 Dataset Understanding and Quality Findings

The raw dataset contains 7,043 customers and 21 columns: customerID, 19 candidate predictors and Churn. Of these, 1,869 (26.54%) churned and 5,174 stayed. IDs were unique and no exact duplicate rows were found. The data dictionary explains the business meaning of every field [3].

The main quality issue involved TotalCharges. Although it represents a number, it was read as text because 11 customers have blank entries. Each blank occurs at zero months of tenure. Labels such as 'No internet service' describe genuine service states and must not be mistaken for missing data.

Member 1 documented these findings in the quality log and passed them to Member 2. This separation matters: identifying a problem in EDA is different from choosing and testing a preprocessing solution.

6.3 Exploratory Findings and Interpretation

EDA insight log extract

Observed pattern	Implication for modelling
Month-to-month churn 42.7%; two-year 2.8%	Retain Contract and interpret it jointly with tenure and services.
Tenure 0–6 months churn 52.9%; 49–72 months 9.5%	Keep tenure continuous; use bands only for description.
Electronic-check churn 45.3%	Preserve payment categories; no causal claim about changing payment.
Fiber-optic churn 41.9%, varying by contract	Compare interaction-sensitive models rather than a single-feature rule.
TechSupport No churn 41.6%; Yes 15.2%	Preserve structural service states and test SupportGap.

These percentages come from all 7,043 historical records. They help generate hypotheses, but they are not evidence that any individual feature causes churn or that a model will generalise to new customers. Source: Member 1's EDA_INSIGHT_LOG.csv.

6.3.1 Contract and Service Interactions

Figure 1 shows that customers on month-to-month contracts churned more often than those on longer contracts. This supports keeping Contract as a predictor, but does not prove that moving someone to a longer contract would prevent churn.

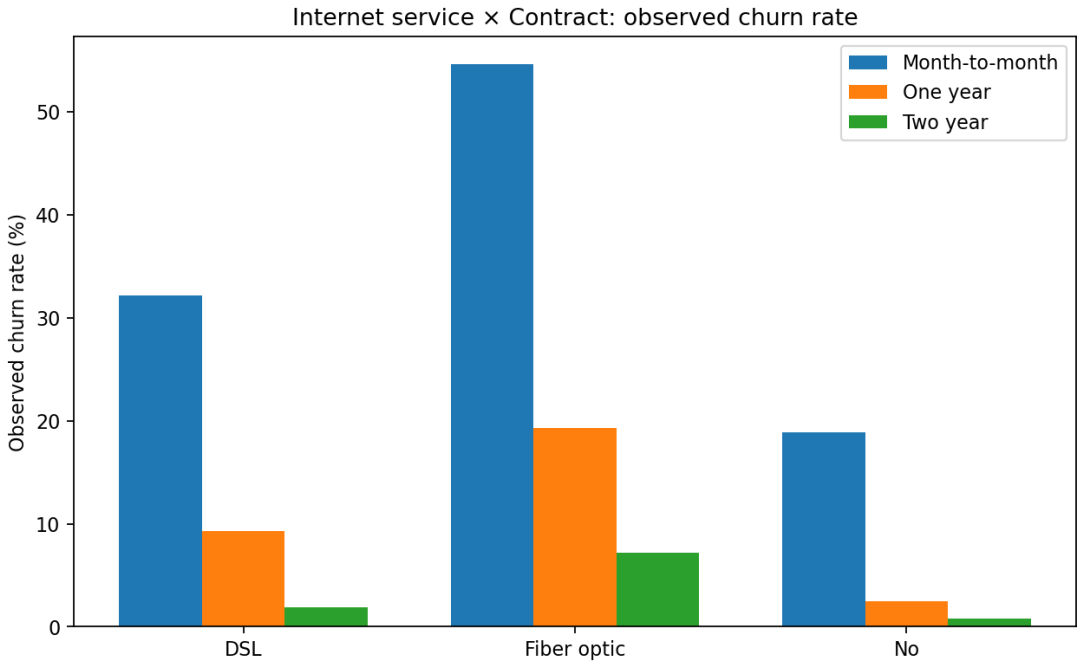


Figure 1. Internet-service and contract interaction.

Churn rates also change when internet service and contract type are considered together. A rule based on 'fiber optic' alone would miss that context. We therefore retained service categories and compared models capable of learning interactions. This is descriptive full-data EDA, not an independent test result.

Source: interaction_internet_contract.png

6.3.2 Tenure and Charge Distributions

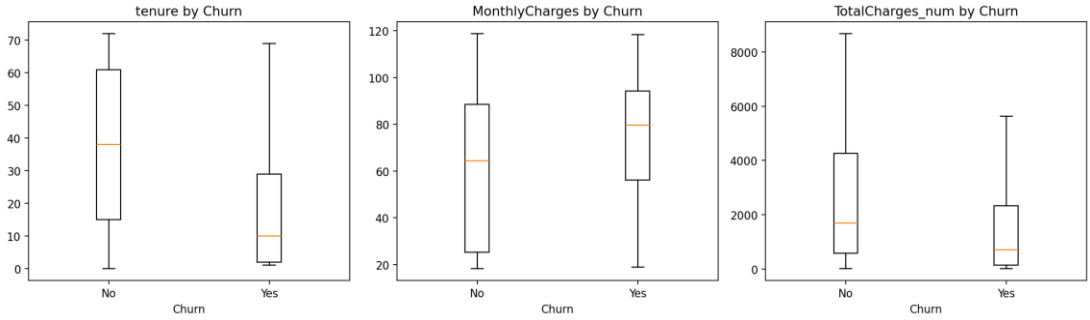


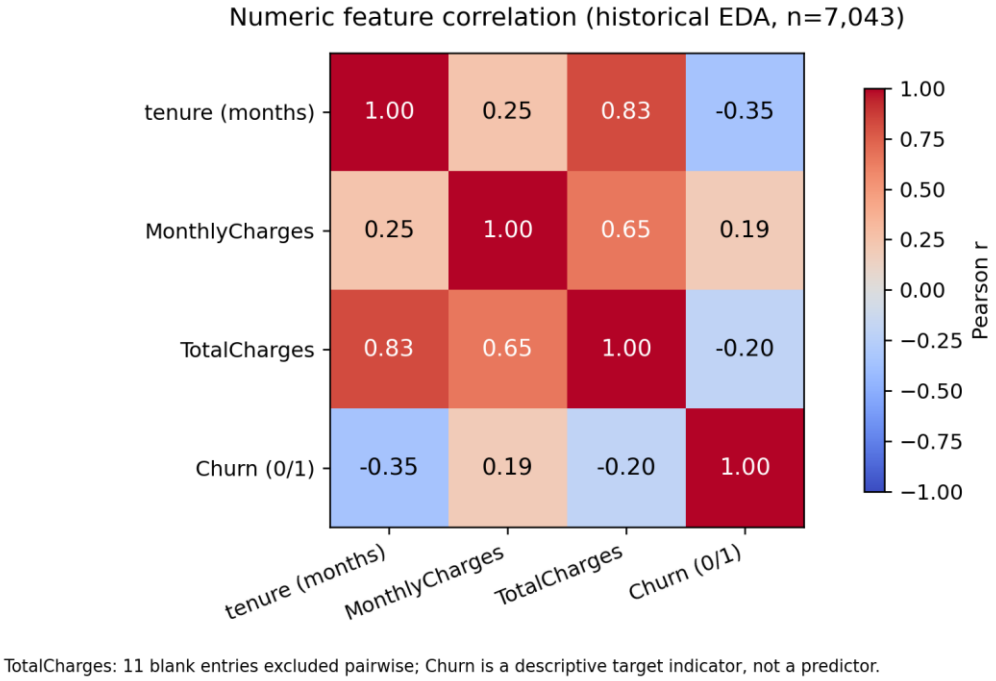
Figure 2. Numeric customer histories by observed churn.

The boxplots compare tenure and charges for customers who stayed or churned. Customers who churned generally had shorter tenure. Because TotalCharges accumulates with time, a low total may simply reflect a short customer relationship, not a cheaper monthly package.

Source: numeric_by_churn_boxplots.png

6.3.3 Supplementary Feature-Association Analysis

We used two heatmaps to understand feature relationships. Pearson correlation measures linear relationships among numeric values; Cramer's V measures the strength of association between categorical values. Both charts use all 7,043 historical records for description only and did not determine the final model panel.



TotalCharges: 11 blank entries excluded pairwise; Churn is a descriptive target indicator, not a predictor.

Figure 2A. Pearson correlations among numeric variables and the binary churn indicator.

Tenure and TotalCharges are strongly positively correlated (r = 0.83), as customers who stay longer usually accumulate more charges. Tenure is negatively correlated with the Churn indicator (r = -0.35), meaning shorter-tenure customers churned more often in this dataset. These are associations, not causes. The 11 blank TotalCharges values were excluded pairwise.

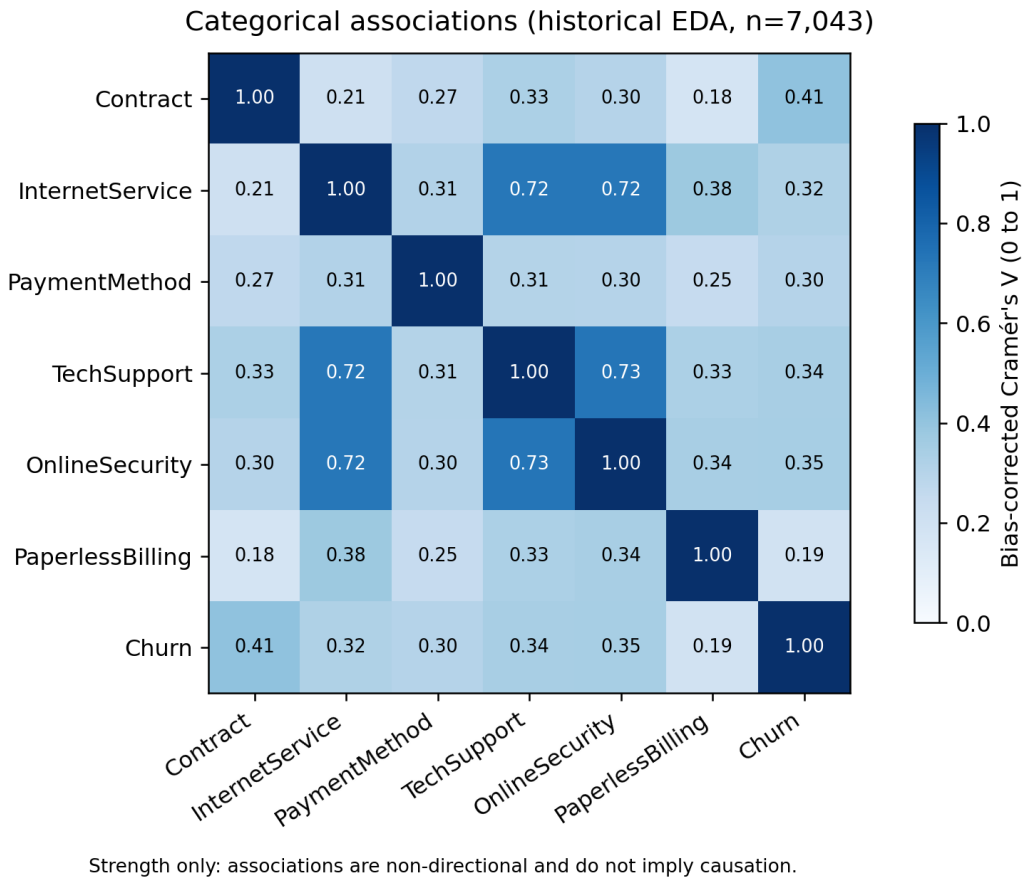


Figure 2B. Bias-corrected Cramer's V for selected categorical features and observed churn.

Of the displayed categorical features, Contract has the strongest association with Churn (Cramer's V = 0.41). OnlineSecurity (0.35), TechSupport (0.34), InternetService (0.32) and PaymentMethod (0.30) also show relationships. Cramer's V measures strength, not direction or causation; shared 'No internet service' states explain part of the relationship between some service fields.

Plot provenance: final_notebook_code/scripts/generate_eda_evaluation_figures.py reproduces the descriptive association diagnostics from dataset/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv. Generated results are under model_method_comparison/results/supplementary_descriptive_visuals/.

6.4 Mixed-Signal Profiles and Descriptive Uncertainty

Member 1 investigated customers with conflicting signals. For example, fiber internet is associated with higher overall churn, but a two-year contract is associated with lower churn. Looking at both features helps avoid oversimplified single-feature rules. We screened subgroups of at least 200 records, although searching many groups can still produce chance patterns.

We added uncertainty estimates rather than treating each churn rate as exact. Wilson intervals describe sampling uncertainty around a subgroup's observed churn rate. A 500-resample bootstrap explores how a contrast changes when the same historical records are resampled. Neither method proves causation or future predictive accuracy.

For instance, roughly 41.9% of fiber-optic customers churned overall, but only about 7.2% of the 429 fiber-optic customers on two-year contracts did. This illustrates why predictors should be interpreted together; the comparison does not demonstrate that contract changes caused the difference.

6.5 Development Disagreement Atlas

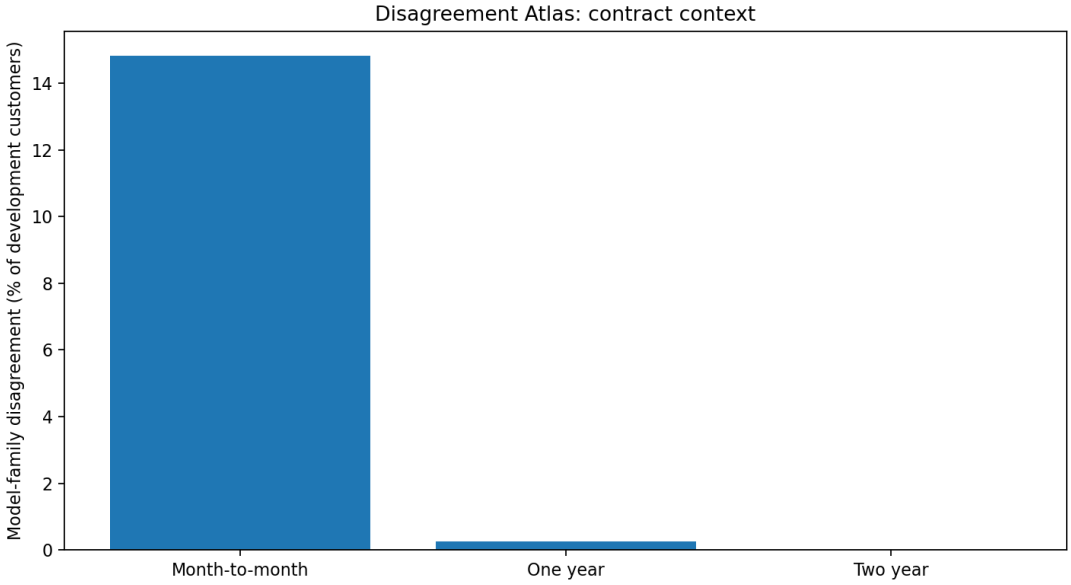


Figure 3. Disagreement Atlas by contract on development customers.

The Disagreement Atlas joins development customer attributes to out-of-fold predictions—predictions made by models that did not train on the same customer. Unlike ordinary churn-rate charts, it asks which customer groups receive different churn/stay labels from otherwise competitive models. High observed churn and high model disagreement are not the same thing.

Source: DISAGREEMENT_ATLAS_CONTRACT.png

Among 3,102 month-to-month development customers, 14.83% had conflicting classifications across the four-family diagnostic. This percentage refers to disagreement between models, not the proportion who churned. It highlights a customer group worth investigating during human review.

6.6 Decision Evidence

Evidence from the log	Reasoning and resulting action
EDA-01 and EDA-02	Contract and tenure patterns support retaining the original predictors. Tenure bands are used for explanation, while continuous tenure is retained for modelling.
EDA-04 and EDA-05	Service interactions motivate nonlinear alternatives and the SupportGap hypothesis. Structural “No internet service” values remain meaningful categories.
EDA-07 to EDA-09	Contradictions, subgroup discovery and bootstrap checks motivate a later audit of customer-level disagreement instead of relying on simple risk stories.
EDA-10	Demographics are passed to data-minimisation and subgroup-error checks; they are not treated as causal levers for retention.

6.7 Lecture Connections, Creative Contribution and Handoff

Lectures 1–2 covered problem framing, variable types, data quality and EDA. We extended those foundations by studying contradictory customer profiles, adding uncertainty estimates and examining customer groups again after model disagreement was measured.

Member 1 passed on the data dictionary, quality and EDA logs, subgroup charts and Disagreement Atlas. These outputs informed Member 2's data preparation, Member 3's modelling questions and Member 4's business interpretation.

7. Member 2 - Preprocessing and Feature Engineering

Imani H.A.I. | IT24102383

7.1 Role and Preprocessing Objectives

Imani prepared the customer data so each algorithm could use it without learning from validation customers. The shared pipeline_contract.py keeps the 19 approved predictors, excludes customerID and Churn from model inputs, and implements the agreed TotalCharges treatment. Using a shared pipeline makes model comparisons more consistent and repeatable.

7.2 Data Cleaning and Input Validation

The shared preprocessing contract checks approved columns, identifiers, split membership and numeric validity. A blank TotalCharges receives zero only at zero tenure; invalid nonblank charges and negative or infinite numeric values are rejected. The new-customer entry point additionally rejects unsupported categories, non-binary SeniorCitizen, fractional tenure and contradictory phone/internet service states. Valid but unusual values are not automatically deleted.

Within the 5,634 development records, tenure and MonthlyCharges are complete and eight TotalCharges values are blank. The recorded IQR check found no values outside its chosen fences for these numeric columns, so no outlier clipping or deletion was applied. Service labels such as 'No internet service' remain valid categories.

7.3 Model-Specific Pipelines and Leakage Prevention

Logistic Regression and SVM use scaled numeric fields and one-hot categorical indicators. The reference Random Forest and Gradient Boosting pipelines use one-hot categories without scaling. CatBoost receives the original categorical representation. The SMOTENC branch scales numeric fields before neighbour selection and samples before one-hot encoding; all learned steps remain inside each training fold.

For example, one-hot encoding changes Contract into separate yes/no indicators for its categories. Scaling makes numerical values more comparable for models affected by feature magnitude. Most tree models do not require that scaling. CatBoost can handle categorical values directly, so we use a different preparation route for it.

The encoder uses handle_unknown="ignore" for a category absent from a fitted training fold. That is different from accepting arbitrary new-customer text: the prediction entry point first checks the supported category domains and service relationships. These checks prevent common input mistakes but do not validate every possible business constraint or prove that an unusual record is realistic.

This fold-by-fold sequence is a key protection against data leakage: fit imputers, encoders, scalers and any oversampling only on each training fold, then apply the fitted steps to the excluded validation fold. Fitting preprocessing once on the full data first would make validation less trustworthy.

7.4 Missing-Value and Imbalance Comparisons

7.4.1 TotalCharges Treatment

Treatment	Mean-fold AP	Interpretation
MEDIAN	0.66143	Comparison; negligible score difference
MEDIAN+IND	0.66164	Comparison; negligible score difference
ZERO	0.66166	Chosen default
ZERO+IND	0.66161	Comparison; negligible score difference

Four methods for the missing TotalCharges entries were compared: median, median plus an indicator, zero, and zero plus an indicator. Their mean-fold AP scores were almost identical. Zero was chosen because the affected customers had zero tenure, making zero cumulative charges a reasonable explanation—not because it greatly improved performance.

7.4.2 Class Imbalance

Strategy	Mean-fold AP	Precision at 0.50	Recall at 0.50
BAL-0	0.66166	0.6529	0.5431
BAL-W	0.65998	0.5169	0.8013
BAL-ROS	0.65875	0.5159	0.8013
SMOTENC	0.64731	0.5346	0.7686

Churn is the minority class (26.54% of the full dataset). The Logistic Regression table compares no balancing (BAL-0), balanced class weights (BAL-W), random oversampling (BAL-ROS) and the completed SMOTENC extension. At probability 0.50, weighting increased recall from 54.31% to 80.13%, while precision fell from 65.29% to 51.69%. More churners were detected, but more non-churners were also flagged.

Random oversampling repeats minority-class examples; SMOTENC constructs synthetic training examples for mixed numeric and categorical inputs. Neither changes the excluded validation records. Logistic Regression without balancing retained the highest AP. The wider comparison below tests the same choice with Random Forest and tuned SVM instead of assuming that one baseline represents every model.

7.4.3 SMOTENC and Model-Specific Imbalance Checks

SMOTENC treats categorical and numeric columns differently. We first repair TotalCharges, impute within the training fold and scale tenure and charges for neighbour distances. SeniorCitizen is categorical in the sampler. Synthetic minority rows are generated before one-hot encoding, with $k = 5$ and seed 42; validation rows are never resampled. Ordinary SMOTE on encoded indicators was deliberately avoided.

Training treatment	LR AP	RF AP	Tuned SVM AP
No balancing	0.66166	0.66386	0.65546
Class weights	0.65998	0.66139	0.65708
Random oversampling	0.65875	0.65716	0.65641
SMOTENC	0.64731	0.64855	0.64615

All values are mean AP across the same five development folds. Each family keeps its model settings fixed while the imbalance treatment changes; this is not an exhaustive joint search. SMOTENC lowered AP for LR, RF and SVM. Class weighting gave SVM its best tested imbalance AP (0.65708), but that remained below the retained tree ensembles. Recall improvements alone did not justify replacing the reference.

The sampler produced 2,115 synthetic rows in each of four training folds and 2,116 in the fifth. All retained legal SeniorCitizen categories, and none violated the tested phone/internet service relationships. This does not prove realistic synthetic customers: interpolated tenure can be fractional and service–charge relationships are not fully constrained. These rows are training constructs only. Results and checks are saved under model_method_comparison/results/classification_revision/.

7.5 Feature Engineering and Data Minimisation

7.5.1 Feature Definitions and Ablation

Constructed feature	Implementation and hypothesis
ServiceCount	Counts phone, additional lines, internet availability and six service add-ons. Tests whether service breadth summarises the customer relationship.
SupportGap	For internet users, counts missing OnlineSecurity, DeviceProtection and TechSupport. Tests whether absence of protective/support services adds information.
AutoPay	Marks automatic bank-transfer or credit-card payment. Tests a compact payment-pattern representation.
ChargePerService	MonthlyCharges divided by max(ServiceCount, 1). Tests a rough charge-per-service measure while preventing division by zero.

Four telecom-inspired features were tested: ServiceCount, SupportGap, AutoPay and ChargePerService. A single-feature ablation adds one feature to an otherwise unchanged baseline. None improved Logistic Regression AP; adding all four to Random Forest also reduced AP. The original 19 predictors therefore remain the historical reference input set.

Feature hypothesis	Change in mean fold AP
ServiceCount — active service count	−0.00008
SupportGap — missing security, protection or support	−0.00012
AutoPay — automatic payment indicator	−0.00015
ChargePerService — MonthlyCharges / max(ServiceCount, 1)	−0.00068
All four together	−0.00068

7.5.2 Demographic Reduction

Input set	Mean-fold AP
All 19 raw predictors	0.66166
Exclude gender	0.66084
Exclude gender and SeniorCitizen	0.66167
Exclude gender, SeniorCitizen, Partner and Dependents	0.66012

Removing gender and SeniorCitizen gave almost unchanged Logistic Regression AP. The nonlinear follow-up improved Random Forest AP from 0.66386 to 0.66547, a small development difference worth further study. This is not evidence of fairness or a validated new policy. The reduced-input model remains a challenger; the full 19-feature reference preserves the recorded panel and test provenance.

7.5.3 Nonlinear Feature Follow-Up

Random Forest input set	Mean-fold AP	Train–validation AP gap
All raw predictors	0.66386	0.09525
Raw plus four engineered features	0.66134	0.11024
Exclude gender and SeniorCitizen	0.66547	0.08929

The engineered set increased the training–validation gap and slightly reduced validation AP. Demographic removal gave a small gain and a smaller gap, so it is a reasonable data-minimisation candidate. It was not silently substituted into the historical panel: a changed panel would require renewed threshold, calibration, disagreement and external validation checks.

7.6 Customer-Level Sensitivity to Imbalance Policy

In the original three-policy Logistic Regression diagnostic, changing among no balancing, class weights and random oversampling altered at least one label for 1,114 of 5,634 customers (19.77%). This used probability 0.50. The count has not been recomputed to include SMOTENC or SVM, and must not be presented as the sensitivity of the expanded experiment. It is separate from TRIAGE voting at 0.35.

7.7 Decision Logs, Lecture Connections and Handoff

The preprocessing log explains the choices for blank charges, structural service categories, encoders, scaling, balancing and engineered features. It also records why some promising ideas were not retained and how validation leakage was avoided.

Lectures 3–4 provided imputation, encoding, scaling, pipelines, feature construction and imbalance handling. We implemented SMOTENC, the mixed-data variant of SMOTE, within training folds and tested whether it improved validation AP. Plain SMOTE after one-hot encoding was not used because interpolated indicators need not represent valid categories. PCA was not needed for this modest, interpretable input space.

Member 2 handed Member 3 the agreed input schema, preprocessing pipelines, feature decisions, fixed folds and sensitivity results. This prevented separate models from being compared using inconsistent versions of the customer data.

8. Member 3 - Models, Tuning and Stability

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8.1 Modelling Objectives and Validation Design

Member 3 used 5,634 development customers and the same five stratified folds to compare models. In each fold, a model learned from four parts and predicted the fifth. Combining these excluded-fold predictions gives each customer an out-of-fold (OOF) score produced without training on that particular record. The 1,409 recorded test customers were not used to retune the final policy in this rerun.

Our main metric was average precision (AP), which evaluates how well a model finds and ranks the minority churn class across different thresholds. We also examined ROC-AUC, precision, recall, F1 and probability error. Mean-fold AP averages five separate fold scores; pooled OOF AP scores all excluded-fold predictions together, so the two numbers can differ.

OOF scores exclude each customer from the model that predicts it, but selecting settings using the same folds still creates selection bias. The original panel comparison retains that limitation. The completed nested SVM evaluation separates inner tuning from outer scoring for that search only. All these analyses are retrospective on an already explored dataset; none restores a pristine historical test.

8.2 Baseline and Alternative Models

Logistic Regression is the baseline. A Decision Tree learns individual splits; Random Forest averages trees; Gradient Boosting adds trees sequentially; and CatBoost handles categorical features natively. The added RBF SVM learns a nonlinear margin after scaling. The table distinguishes the original candidates from the later SVM extension, rather than implying that every model belonged to the historical panel.

Model	Mean fold AP	Fold SD	Pooled OOF AP
Logistic Regression	0.6617	0.0216	0.6585
Decision Tree	0.6140	0.0258	0.6190
Random Forest	0.6575	0.0240	0.6539
Gradient Boosting	0.6722	0.0233	0.6685
Tuned Random Forest	0.6639	0.0205	0.6605
CatBoost	0.6691	0.0240	0.6648
Tuned CatBoost	0.6727	0.0240	0.6693
Tuned SVM (later extension)	0.6555	0.0152	0.6527

8.3 Hyperparameter Tuning and Comparison

Model	Search and retained setting	Mean-fold AP outcome
Logistic Regression	C = 0.01, 0.1, 1, 10, 100; retain C = 1.	0.66166; baseline retained

Model	Search and retained setting	Mean-fold AP outcome
Random Forest	Depth 5, 10, unrestricted; leaf minimum 5 or 10; 200 trees fixed. Retain depth 10, leaf minimum 10.	0.65746 initial → 0.66386 tuned
Gradient Boosting	Learning rate 0.03, 0.05, 0.10; depth 2 or 3; 100 estimators fixed. Retain 0.05 and depth 3.	0.67218; initial configuration retained
CatBoost	Depth 4 or 5; learning rate 0.03 or 0.05; 500 iterations fixed. Retain depth 4 and 0.03.	0.66912 initial → 0.67273 tuned
RBF SVM (extension)	C = 0.1, 1, 10; gamma = scale, 0.01, 0.1. Retain C = 1 and gamma = 0.01.	0.63740 default → 0.65546 tuned; nested outer AP also 0.65546

Tuned CatBoost had the best mean-fold AP (0.67273), only 0.00055 above Gradient Boosting (0.67218). This small difference is not convincing proof that CatBoost is superior in general. Furthermore, scores used for both tuning and selection may be somewhat optimistic because the folds were reused.

Hyperparameters are settings chosen before training, such as how deep trees may grow, how many examples a leaf must contain, the boosting learning rate or Logistic Regression regularisation strength. We tested recorded alternatives. The retained Random Forest, for example, limits depth to 10 and requires at least 10 samples per leaf to control complexity.

Fold-score variability tells us how much overall model performance changed across different validation subsets. Prediction disagreement is different: it asks whether two models classify the same customer differently. Our project investigates both.

8.3.1 Completed SVM Search

The nine-setting grid contains 45 training/validation fits across five saved folds, followed by refitting the selected pipeline. C controls the penalty for training errors; gamma controls how locally an RBF kernel responds to nearby records. We selected by AP rather than training accuracy. The chosen C = 1, gamma = 0.01 improved ranking relative to the earlier default SVM. Its decision_function output is a margin, not a calibrated probability; a 0.35 probability threshold cannot be applied to it.

8.4 Generalization, Regularization and Learning Curves

To check overfitting, we compared performance on each model's training folds with its excluded validation folds. Tuned Random Forest had the largest mean AP gap (0.0953), compared with 0.0469 for Gradient Boosting, 0.0372 for CatBoost and 0.0055 for Logistic Regression. The forest therefore deserves extra caution, even though its validation AP was still competitive.

L1 and L2 Logistic Regression both achieved about 0.6617 validation AP. The earlier untuned RBF SVM achieved AP 0.6374 and a training–validation gap of 0.0749. After tuning C and gamma, SVM reached AP 0.6555 with training AP 0.6639 and a smaller gap of 0.0085. This is evidence that the tested settings reduced training-specific fit, not proof that SVM cannot overfit.

The learning curves, regularisation checks and SVM extension are development-only diagnostics completed after the operational panel was recorded. The default-SVM result in Figure 4 is retained as an

earlier reference. The tuned SVM and nested validation results below describe the completed extension; they did not retrospectively alter the historical voting policy.

A learning curve compares training and validation scores at different training-set sizes. In our sample, Random Forest's validation AP improved with more training records while its training score stayed higher. This shows both useful learning and remaining training-specific fit; improvement on future datasets is not guaranteed.

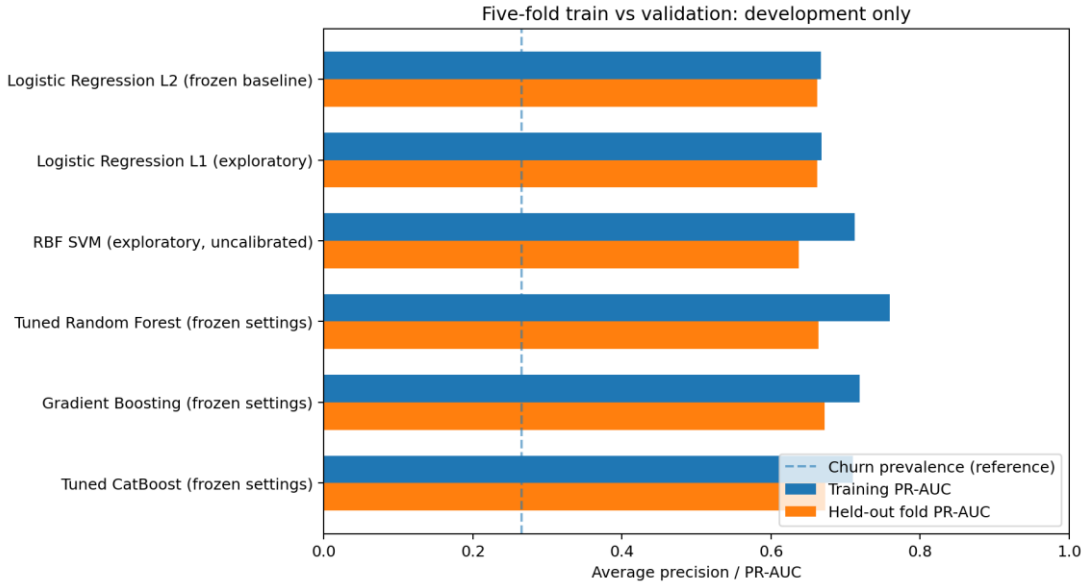


Figure 4. Training and validation AP on development folds.

The chart compares the original generalisation audit, including the earlier untuned SVM. Training AP measures fitted records; validation AP measures excluded records. Random Forest has the largest gap. The newer tuned-SVM gap is 0.0085, reported separately below. A small gap alone does not establish useful predictive performance or guarantee future generalisation.

Source: train_vs_validation_pr_auc.png

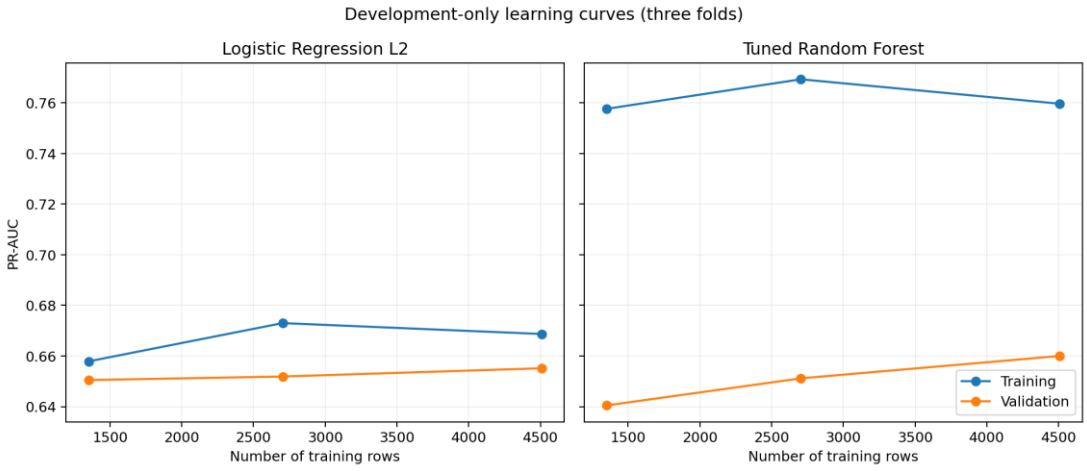


Figure 5. Learning curves for Logistic Regression and Tuned Random Forest.

The curves show how Logistic Regression and tuned Random Forest behaved as training size increased. Forest validation AP improved but stayed below its training AP. These three-fold exploratory curves should not be confused with the main five-fold model-selection scores.

Source: development_learning_curves.png

8.4.1 Nested SVM Validation

For each saved outer fold, we hold that fold aside, tune the same nine SVM settings with three stratified inner folds using only the outer training rows, refit the winner on those training rows and score the excluded outer rows. Scaling and encoding are re-fitted inside every inner pipeline. This prevents that outer fold from choosing C or gamma.

Outer fold	Selected C	Selected gamma	Training AP	Outer AP
1	1	0.01	0.6650	0.6458
2	1	0.01	0.6701	0.6365
3	1	0.01	0.6607	0.6731
4	1	0.01	0.6598	0.6679
5	1	0.01	0.6639	0.6539

All five outer folds selected C = 1 and gamma = 0.01. Mean outer AP was 0.65546 (fold SD 0.01517), mean training AP was 0.66392 and the mean gap was 0.00846. The nested and ordinary five-fold results match because every inner search selected the same settings as the full development search; this equality is not assumed in advance.

These results assess the SVM search procedure, not the entire project's model-selection process. A small or even negative fold gap can occur through sampling variation. We interpret gaps with validation performance, fold variability and learning curves rather than claiming that overfitting has been eliminated. The historical test remains previously exposed.

8.5 Near-Optimal Model-Panel Selection

The historical panel contains the original candidates with mean-fold AP no more than 0.010 below the best: CatBoost, Gradient Boosting and tuned Random Forest. The initial proposal's example 2% ROC-AUC rule was changed after development inspection and recorded as an exploratory decision. The later tuned SVM's AP is 0.01727 below CatBoost and its margins are not probabilities, so it is not added to the 0.35 vote rule.

Candidate	Mean-fold AP	Gap from best	Within 0.010
Tuned CatBoost	0.67273	0.00000	Yes
Gradient Boosting	0.67218	0.00055	Yes
Tuned Random Forest	0.66386	0.00888	Yes
Logistic Regression	0.66166	0.01107	No
Decision Tree	0.61395	0.05878	No

A tighter 0.005 AP band keeps only CatBoost and Gradient Boosting; 0.010 adds tuned Random Forest; 0.015 also includes Logistic Regression. These comparisons show that panel membership depends on the tolerance we choose. 'Near-optimal' describes this rule, not statistical equivalence.

8.6 Four Sources of Prediction Instability

Development diagnostic at 0.50	Customers affected	Share
Imbalance-policy preprocessing sensitivity	1,114	19.77%
Four model families	463	8.22%
Accepted CatBoost configurations	158	2.80%
RF and CatBoost seed sensitivity combined	233	4.14%

We ran four development diagnostics to ask why the same customer can receive different churn/stay labels. The groups overlap, so their percentages must not be added. The four-family diagnostic includes Logistic Regression as well as the three operational-panel families and uses a diagnostic 0.50 cutoff.

8.6.1 Model Family

Across the four families, 463 of 5,634 development customers (8.22%) received at least one conflicting classification at threshold 0.50. Thus, models can have similar overall ranking scores but disagree about particular individuals.

8.6.2 Accepted Configuration

Keeping CatBoost as the model family but changing among three similarly scoring settings led to different classifications for 158 customers (2.80%). This isolates the effects of the tested hyperparameters.

8.6.3 Random Seed

Changing random seeds to 42, 43 and 44 for selected Random Forest and CatBoost configurations affected 233 customers (4.14%) across the combined diagnostic. This shows sensitivity within those three seeds, not every possible training run.

8.6.4 Imbalance Policy and Overlap

The original three-policy Logistic Regression screen found 1,114 sensitive customers. Some also appear among the 463 family-sensitive customers. These historical diagnostic counts refer to their stated candidate sets; they exclude the later SMOTENC and SVM extension and must not be treated as an updated six-family sensitivity analysis.

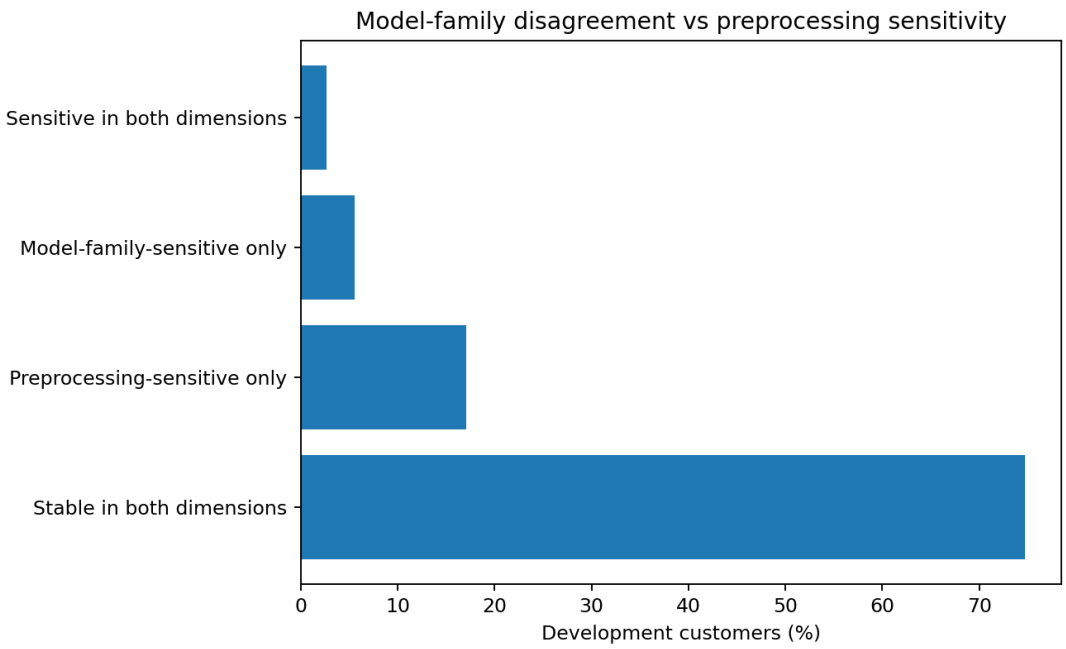


Figure 6. Combined preprocessing and model-family sensitivity.

The overlap chart shows 4,208 development customers stable under both tested changes, 963 sensitive only to preprocessing policy, 312 sensitive only to model family and 151 sensitive to both. It demonstrates that different modelling choices can affect different people. All counts use the diagnostic 0.50 threshold.

Source: robustness_quadrants.png

For a customer with predictions [1, 0, 0], two model pairs disagree, but only one customer is affected. We therefore distinguish 'number of disagreeing pairs' from 'number of customers with any disagreement' instead of treating those as the same result.

Even unanimous models can share the same mistake because they are trained on related data. Our vote count describes agreement at a chosen threshold, not a guaranteed probability that the result is correct.

8.7 Decision Logs, Lecture Connections and Handoff

Decision logs D07–D08 record the original comparison and panel. Revision records R04–R06 add SVM tuning, overfitting evidence and nonlinear feature comparisons. The original tolerance remains exploratory. Nested SVM validation supports an assessment of that tuning procedure, not statistical equivalence among the operational models.

Lectures 5–7 informed classification, validation, regularisation, bagging, boosting and tuning. The completed extension explicitly tests SVM C and gamma and uses nested validation to separate its tuning from outer scoring. Member 1 uses the original OOF disagreements for the Atlas; Member 4 uses the retained probability panel for threshold and business evaluation.

9. Member 4 - Evaluation and Business Decisions

H.A.S.R. Jayawardhana | IT24102243

Member 4 asked whether these predictions would help a telecom company with limited contact capacity. Jayawardhana evaluated model errors, probability quality, subgroup differences and review workload, then compared the four-state TRIAGE policy with a simpler risk-ranking approach.

9.1 Evaluation Objectives and Metrics

Metric or diagnostic	Meaning for this project
Average precision (AP)	Summarises the precision–recall ranking over thresholds. Useful because churn is the minority outcome. Files sometimes label it PR-AUC; the reported implementation uses average_precision_score.
Precision and recall	Precision asks how many flagged customers actually churned; recall asks how many churners were found. Increasing recall can require contacting more non-churners.
ROC-AUC	Measures ranking of churners above non-churners. It complements AP but does not define the size of a contact queue.
F1 and confusion counts	F1 balances precision and recall at one threshold. False negatives are missed churners; false positives are contacts to recorded non-churners.
Brier score and reliability curve	Check probability error and whether observed frequencies resemble predicted probabilities. They do not prove perfect calibration.
Capture at a fixed budget	Asks how many known churners appear in the top fraction of customers that the team can contact.

9.2 Threshold Selection and Calibration

The evaluation included different decision thresholds because the cutoff changes how many churners we detect and how many customers who would stay are mistakenly flagged. We used development OOF predictions to compare these costs, not future outcomes of actual retention calls.

A model votes 'churn' when its estimated probability is at least 0.35. Lowering the threshold can find more churners but also contact more people who would stay. We compared this trade-off on development

predictions; without real contact costs or offer-response data, 0.35 is an operating choice, not a proven financial optimum.

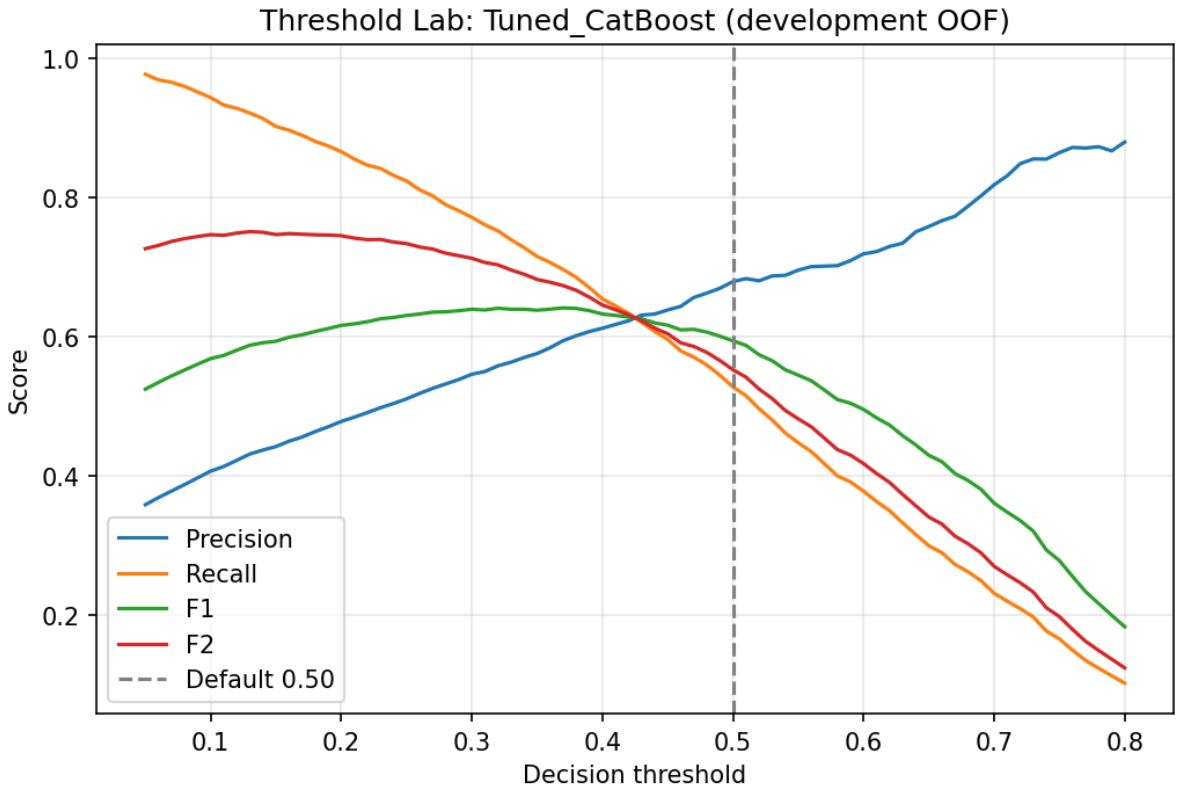


Figure 7. Threshold trade-offs in development predictions.

Figure 7 shows precision, recall and F-scores at different cutoffs. The dashed 0.50 reference is not our final policy; our recorded 0.35 cutoff favours finding more churners, while accepting more false alarms. This analysis does not show that making a retention offer prevents churn.

Source: threshold_lab_plot.png

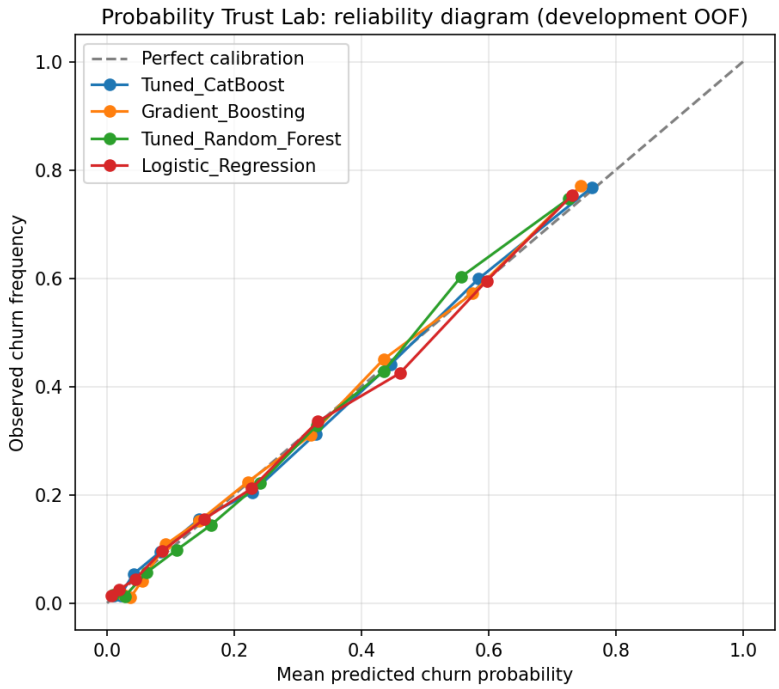


Figure 8. Reliability diagram from the calibration analysis.

The reliability chart groups predictions by their estimated risk and compares them with observed churn frequencies. It helps check whether a predicted probability is meaningful. We examined calibration but did not fit an additional calibrator for the final policy.

Source: reliability_diagram.png

CatBoost threshold	Precision	Recall	Customers flagged
0.35	0.5757	0.7151	1857
0.50	0.6790	0.5278	1162

For CatBoost alone on development OOF predictions, moving from 0.50 to 0.35 increased recall from 0.5278 to 0.7151, while reducing precision from 0.6790 to 0.5757. It flagged 1,857 rather than 1,162 people. The final ACT group is smaller because all three models must vote positive.

The new imbalance comparison uses probability 0.50 for LR/RF and margin zero for SVM when reporting precision and recall. Those are diagnostic operating points. Resampling can change probability calibration; none of the new branches inherits the historical 0.35 policy automatically. Adding SVM would require validated probability calibration and a new policy evaluation.

9.3 Final TRIAGE Decision Rule

Positive votes at probability ≥ 0.35	State	Interpretation
3 of 3	ACT	Prioritise for a reviewed retention contact
2 of 3	REVIEW	Positive majority with disagreement
1 of 3	MONITOR	Negative majority with disagreement
0 of 3	HOLD	Lower priority, with no guarantee of staying

Under the final decision rule, each of three models votes positive at probability 0.35 or higher. Three votes produce ACT, two REVIEW, one MONITOR and none HOLD. The 0.50 threshold elsewhere in the report was used only for separate sensitivity diagnostics. Models can agree and still be wrong.

Among 1,677 development ACT customers, 1,676 had month-to-month contracts and 1,337 used fiber internet. Their median tenure was eight months, compared with 29 months across development customers. These profiles show whom the rule identifies, not which retention treatment would work.

9.4 Historical-Test Performance and Error Analysis

The historical test includes 1,409 customers, of whom 374 churned. We replayed the previously selected models and 0.35 threshold rather than changing them in response to these test outcomes. However, some test information had already been viewed during earlier analysis, so these scores are not a pristine independent test.

Model	AP	ROC-AUC	Precision	Recall	F1
Tuned CatBoost	0.6658	0.8459	0.5739	0.7166	0.6373
Gradient Boosting	0.6607	0.8453	0.5711	0.7086	0.6325
Tuned Random Forest	0.6547	0.8434	0.5702	0.7273	0.6392

At threshold 0.35, CatBoost correctly detected 268 churners and correctly left 836 non-churners unflagged. It missed 106 churners and flagged 199 non-churners. Its Brier score was 0.1349. A bootstrap interval for AP was 0.6187–0.7143, but this resamples fixed predictions and does not include the effects of retraining or model selection.

Triage state	Customers	Share	Observed churn
ACT	431	30.59%	58.93%
REVIEW	43	3.05%	44.19%
MONITOR	29	2.06%	17.24%
HOLD	906	64.30%	10.60%

Of the 374 historical churners, 254 fell in ACT (67.91% ACT-only recall). HOLD still contained 96 churners, so it must not be interpreted as 'safe'. REVIEW and MONITOR contained 72 customers together (5.11% of the test set), giving a possible estimate of cases needing extra staff review.

Do not confuse a single-model confusion matrix with TRIAGE's four-state results. CatBoost flagged 467 test customers on its own; ACT requires positive votes from all three models, so it includes fewer customers and has a different recall.

REVIEW and MONITOR indicate split votes, not automatic binary errors. They should be interpreted using their observed churn rates or a separately specified final action rule, rather than silently counting them as false positives or negatives.

9.4.1 Visual Diagnostics for the Recorded Historical Test

The ROC, precision–recall and confusion-matrix charts below were reproduced from saved probabilities for the 1,409 historical-test customers. No model was retrained to draw them. They agree with the recorded evaluation scorecard, but prior test exposure still limits independent-validation claims.

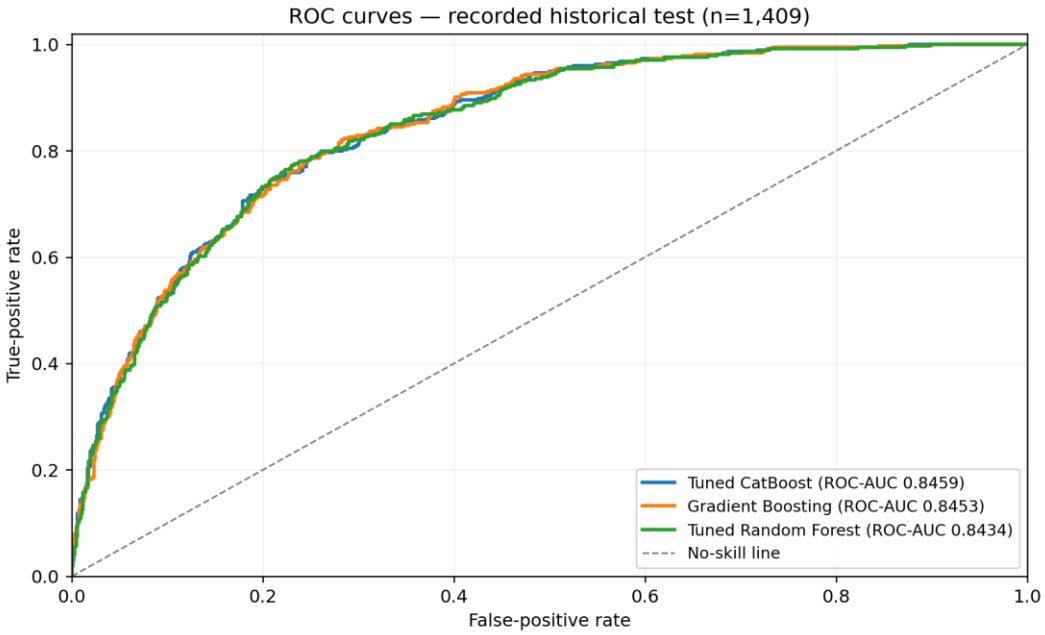


Figure 8A. ROC curves for the three frozen candidate models on the recorded historical test.

The ROC curves are nearly identical: CatBoost ROC-AUC = 0.8459, Gradient Boosting = 0.8453 and Random Forest = 0.8434. ROC-AUC measures how well churners are generally ranked ahead of non-churners across possible thresholds. Differences this small do not establish a decisive winner.

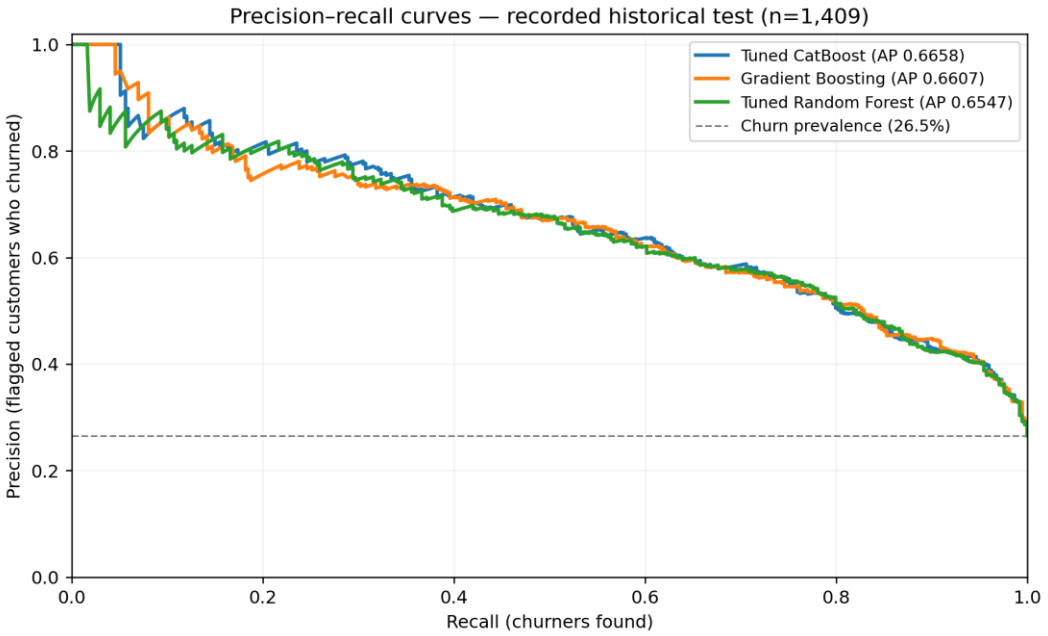


Figure 8B. Precision-recall curves for the three frozen models on the recorded historical test.

The precision–recall curves show the trade-off between finding churners and generating false alarms. Recorded AP was 0.6658 for CatBoost, 0.6607 for Gradient Boosting and 0.6547 for Random Forest, compared with roughly 26.5% churn prevalence. AP does not directly tell us how many customers belong in ACT.

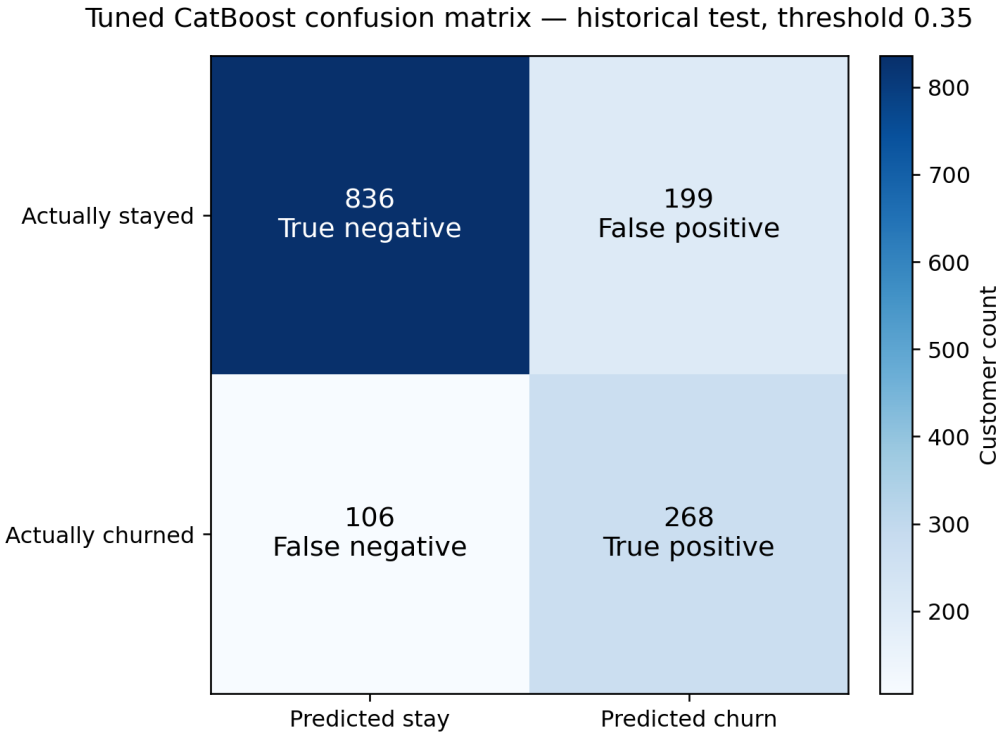


Figure 8C. Tuned CatBoost confusion matrix at the recorded 0.35 threshold.

The CatBoost confusion matrix separates correctly identified churners (268) from missed churners (106), false alarms (199) and correctly unflagged non-churners (836). The four-state TRIAGE system is evaluated separately because it does not directly predict only Yes or No.

Plot provenance: archived model_method_comparison/results/member4/final_test_probabilities_20261002_100144.csv. The included final_notebook_code/scripts/generate_eda_evaluation_figures.py recreates the historical ROC/PR and confusion-matrix diagnostics without fitting a new model.

9.5 Selective Coverage and Review Workload

We also examined a selective-review idea: issue a definite vote-based decision only when models agree and leave split-vote cases for staff review. This introduces a trade-off between how often the method makes a decision (coverage) and how correct those decisions are.

Coverage is the percentage of cases decided without abstaining. Correctness may look better if harder customers are excluded, so it must always be reported alongside coverage. Saved results are available in the Member 4 results folder on GitHub.

REVIEW and MONITOR together contain 72 recorded test customers (5.11%). This helps estimate potential review workload, but does not prove that extra review would reduce churn.

9.6 Contact-Budget Comparison

Because retention teams have limited capacity, we compared two ways to choose whom to contact: rank customers by the three models' average probability, or use TRIAGE's agreement-aware priority. Both methods used the same recorded predictions and the same contact limits.

Contact budget	Customers	Risk-only capture	TRIAGE capture
10%	140	28.34%	28.34%
20%	281	50.53%	50.53%
30%	422	66.58%	66.84%

At 10% and 20% contact capacity, the two methods found the same number of historical churners. At 30%, TRIAGE found just one additional churner. Development resampling did not establish a general gain, and in one comparison TRIAGE performed worse than ranking by CatBoost alone. Capturing a historical churner is not evidence of preventing churn.

For example, at 20% capacity each method reached 189 of 374 historical churners within 281 contact slots. At 30%, ordinary panel-average ranking reached 249 while TRIAGE reached 250 within 422 slots. A one-person difference does not demonstrate meaningful business improvement.

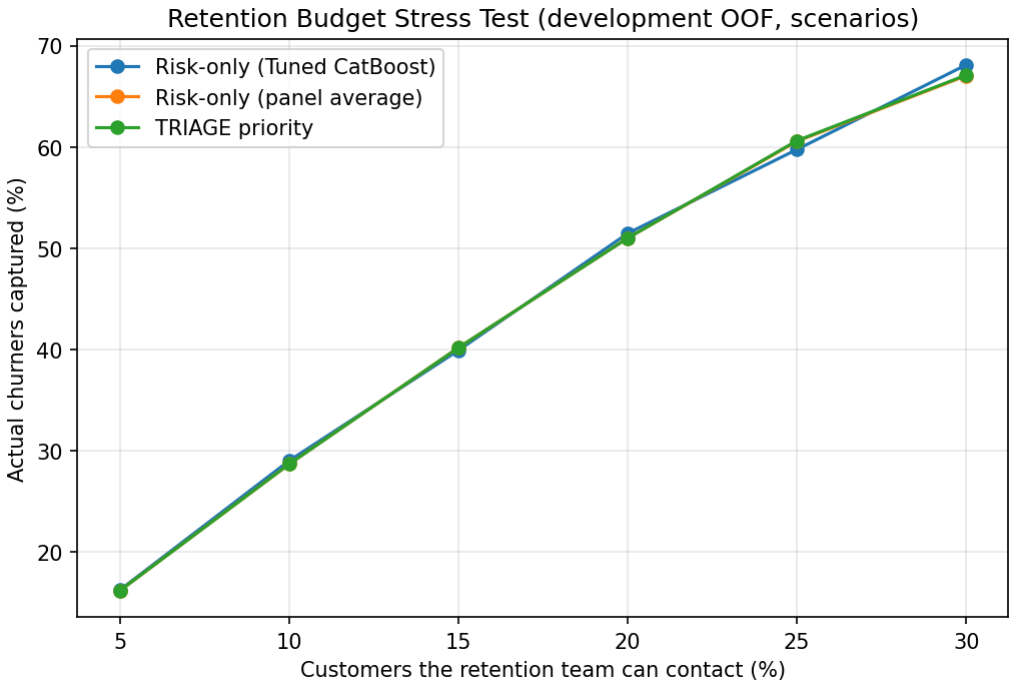


Figure 9. Development contact-budget stress test.

Figure 9 shows almost overlapping development capture curves. This negative finding matters: a more complicated prioritisation rule did not reliably outperform basic risk ranking. The chart uses development OOF predictions; the table above gives the separate historical-test replay.

Source: budget_stress_plot.png

9.7 Subgroup Findings and Limitations

The subgroup audit found unequal error patterns. In the historical test, ACT-only recall was 66.84% for women and 69.06% for men. Senior customers had higher recall but also a higher false-positive rate (37.90% versus 14.27%). None of nine churners with two-year contracts entered ACT; this small sample remains uncertain but important.

Development data also showed zero ACT-only recall among one- and two-year contract churners. These are different results from the historical test and should not be mixed. The project does not prove that predictions are fair across demographic or contract groups.

A higher subgroup churn prevalence does not by itself explain a higher false-positive rate: FPR is measured among actual non-churners. The audit describes observed differences without assigning their cause. Likewise, overlapping individual confidence intervals do not establish model equivalence; paired comparisons and practical performance differences must be distinguished.

This remains a research and decision-support prototype. The data are historical snapshots, not intervention outcomes. Original tuning reused development folds and the test was previously exposed. Nested SVM validation addresses inner parameter selection for one challenger only; it does not validate all earlier choices or establish performance on future customers.

9.8 Decisions, Lecture Connections and Final Recommendation

Decision records D09–D13 explain the 0.35 cutoff, the four vote states, agreement limitations and contact-capacity results. The bootstrap intervals describe uncertainty in the saved predictions, not variation from retraining models or choosing new settings.

Lectures 5–7 supplied the foundations for classification metrics, cross-validation, ensembles and calibration. We extended those ideas by asking whether disagreement helps a real retention team decide what to contact or review. Subgroup errors and capacity tests shaped the recommendation.

We recommend a limited, human-reviewed pilot: prioritise by estimated churn risk, show disagreement to reviewers, record outreach costs and customer outcomes, and compare against a suitable baseline. Without an actual comparison of contacted and uncontacted customers, we cannot claim that the model prevents churn.

10. Project Integration and Creative Contributions

10.1 Integration of the Four Members' Contributions

Stage and owner	Output passed to the next stage
Member 1 to Member 2	Data dictionary, quality issues, EDA insights and business hypotheses.
Member 2 to Member 3	Model-aware pipeline contract, feature decisions, saved folds and preprocessing sensitivity.
Member 3 back to Member 1	Customer-level OOF disagreement for the Disagreement Atlas.
Members 1 and 3 to Member 4	Customer profiles, model probabilities, stability diagnostics and panel definitions.
Member 4 to the group	Threshold and triage interpretation, test replay, capacity comparisons, subgroup risks and final recommendation.

The key connection between members is a feedback loop. EDA identified business and data questions; preprocessing created a consistent input contract; models exposed disagreements; and EDA examined which customers were affected. Member 4 then tested whether this extra information improved decisions. The project is one joined workflow, not four unrelated notebooks.

Customer IDs align dataset rows, OOF probabilities and stability flags across the member handoffs. This prevents accidental mismatches if row order changes. Sections 6–9 provide the detailed evidence.

10.2 Project-Specific Contributions and Supporting Evidence

Contribution	Purpose and evidence	Boundary
Four-state decision layer	Makes three-model agreement visible; operational definitions and results in Section 9.3–9.4.	A deterministic policy, not a new learning algorithm or correctness guarantee.
Disagreement Atlas	Returns OOF disagreement to customer profiles; Section 6.5.	Exploratory descriptions, not causal rules.
Separate sensitivity analyses	Distinguishes model family, configuration, seed and imbalance-policy effects; Section 8.6.	Only the recorded choices are tested; affected groups overlap.
Capacity-based evaluation	Compares TRIAGE with simpler ranking at equal contact capacity; Section 9.6.	No reliable improvement in capture or measured retention benefit.

The contribution is the combined decision-support workflow, not a newly invented algorithm. It joins profiling, sensitivity analysis, validation and a transparent vote rule. We also report negative evidence: engineered features did not improve the tested LR or RF reference, SMOTENC lowered AP in all three tested classifier families, and TRIAGE did not reliably beat simpler risk ranking.

10.3 Connection to Lecture Concepts and Scope Choices

Lecture	**Applied concept and evidence**
1 — Introduction	Business framing, supervised task and data-to-decision workflow; Sections 1–5 and 6.1.
2 — Data understanding	Types, quality checks, descriptive EDA and interaction analysis; Sections 6.2–6.4.
3 — Preprocessing	Imputation, encoding, scaling and fold-fitted pipelines; Sections 7.2–7.3.
4 — Feature engineering	Feature construction, linear/nonlinear ablations, demographic reduction and fold-local SMOTENC; Sections 7.4–7.6.
5 — Classification	Classifier comparison, tuned RBF SVM and classification metrics; Sections 8.2–8.4 and 9.1.
6 — Regression concepts adapted	Generalisation, learning curves and L1/L2 concepts adapted to binary classification; nested SVM validation in Section 8.4.1.
7 — Ensembles and improvement	Bagging, boosting, grid search, complexity control and calibration diagnostics; Sections 8.2–8.4 and 9.2. No extra calibrator was fitted.

We selected methods relevant to binary churn and documented alternatives. SMOTENC and tuned SVM are implemented and evaluated; kNN, Naïve Bayes, PCA and stacking are not. The assignment requires defensible choices, not every algorithm from the slides. Averaging probabilities is not stacking, and Lecture 6 regularisation concepts were adapted to classification rather than changing the target to regression.

11. What Happens When a New Customer Is Entered

11.1 Required Input, Setup and Validation

The prototype includes predict_new_customers.py for scoring new records. First, rebuild_frozen_models.py reconstructs the fitted model objects from saved development data and settings. The prediction script then loads those objects and scores incoming customers without retraining on them. This is a Python prototype, not a live website or API.

A new customer must supply the 19 approved predictor fields. The Churn target cannot be an input because that is what the model predicts. customerID may be supplied to label the output, but is not passed into the models. Field names and numeric values must follow the documented input contract.

The script checks required columns, unexpected fields and unique nonblank batch IDs. It rejects negative or infinite numeric values, SeniorCitizen outside 0/1, fractional tenure and unsupported category labels. PhoneService must agree with MultipleLines, and InternetService must agree with each internet add-on's structural category. Blank TotalCharges is permitted only at zero tenure. No arbitrary maximum charge is imposed.

Step	Processing and output
1. Validate	Validate schema and IDs; reject target leakage, invalid numeric values, unsupported categories, non-binary senior status, fractional tenure and contradictory service states. Blank TotalCharges requires zero tenure.
2. Transform	Use the fitted tree pipelines for Random Forest and Gradient Boosting. Use the saved native preparation object before CatBoost. Do not fit an encoder or scaler on the incoming customer.
3. Score	Each of the three fitted models returns a churn probability. A number such as 0.60 is a model score, not a certain prediction that churn will occur.
4. Vote	For each model, probability ≥ 0.35 gives a positive vote. Sum the three votes.
5. Assign state	3 → ACT; 2 → REVIEW; 1 → MONITOR; 0 → HOLD. Retain individual probabilities so disagreement is visible.
6. Support a decision	Show the result to a retention reviewer. Rank by risk when capacity is limited; an ACT label does not prove an offer will prevent churn.

To score a customer, the program validates inputs, applies the fitted transformations, calculates churn probabilities from all three models, counts votes at 0.35 and assigns ACT, REVIEW, MONITOR or HOLD. These stages allow staff to trace the final state back to the model outputs.

11.2 Executed Synthetic Examples

The synthetic examples below were actually scored by the reconstructed models. They are artificial customer records with no observed future churn outcomes, so they demonstrate the code pathway—not accuracy or retention benefit.

Input field	Synthetic A	Synthetic B
gender	Female	Female
SeniorCitizen	0	0
Partner	No	Yes
Dependents	No	Yes
tenure	4	60
PhoneService	Yes	Yes
MultipleLines	No	No
InternetService	Fiber optic	DSL
OnlineSecurity	No	Yes
OnlineBackup	No	Yes
DeviceProtection	No	Yes
TechSupport	No	Yes
StreamingTV	Yes	Yes
StreamingMovies	Yes	Yes
Contract	Month-to-month	Two year
PaperlessBilling	Yes	Yes
PaymentMethod	Electronic check	Bank transfer (automatic)
MonthlyCharges	90.0	75.0
TotalCharges	360.0	4500.0

Profile	CatBoost	Gradient boosting	Random forest	Votes / state
Synthetic A	0.7430	0.7331	0.7387	3 / ACT
Synthetic B	0.0211	0.0280	0.0153	0 / HOLD
Synthetic C	0.9210	0.8572	0.8306	3 / ACT

Synthetic A gets three positive votes and ACT; Synthetic B gets zero votes and HOLD. Synthetic C has the same values as A except zero tenure and blank TotalCharges, which correctly follows the structural-zero preparation rule and also receives ACT. Differences in model scores should not be attributed to any one changed field without a separate explanation method.

As a negative test, an artificial customer with four months of tenure and blank TotalCharges is rejected. The code reports: 'Structural-zero treatment requires tenure=0 for missing TotalCharges'. This guards against treating every blank charge as a genuine zero.

The sample inputs are in dataset/examples/synthetic_customers.csv, the scoring code in final_notebook_code/predict_new_customers.py, and saved example outputs in model_method_comparison/results/new_customer_demo/. The repository allows a reviewer to inspect both the inputs and recorded scores.

11.3 Split Votes and Limits of the Output

The synthetic examples demonstrate ACT, HOLD and a rejected blank-charge record; they do not demonstrate all four states. A separate validation check accepts the development inputs and rejects seven malformed cases, covering invalid senior status, tenure, category, numeric values and service contradictions. Synthetic examples and input checks are software evidence, not prediction-accuracy estimates.

For a new customer we do not yet know the actual churn outcome, so precision, recall, F1 and confusion counts cannot be calculated for that customer. Models may all agree and still be wrong, and no method here estimates whether an offer would persuade the customer to stay.

Before live deployment, the team still needs broader business plausibility checks, secure artifact handling, versioned scoring logs, monitoring for changing customer populations and measured retention outcomes. Supported categories and basic service-consistency checks are now implemented; these should not be listed as entirely unimplemented future work.

12. Decision Logs and Assignment Evidence

12.1 Consolidated Decision Log

D01–D14 preserve the original project decisions. The completed extension adds R01–R12 in decision_logs/project/classification_revision_decisions.csv. The tables below summarise both without rewriting historical choices. Detailed ownership and original evidence paths remain in CONSOLIDATED_DECISION_LOG.csv and the member logs.

ID	Alternatives or question	Evidence and recorded action
D01	Overwrite raw data or preserve it	Preserve the raw file; immutable fingerprints support traceability. Section 6.2.
D02	Median, zero, with/without indicator	Negligible AP differences and blanks at zero tenure support zero treatment. Section 7.4.1.
D03	Use ID/derived outcome fields or exclude	Exclude identifier and target-derived inputs; retain ID only for joins. Sections 4.2 and 7.2.
D04	No balancing, weights, ROS or SMOTENC	Original LR choice retained. The extension tests LR/RF/SVM and finds lower AP with SMOTENC in each family. Section 7.4.
D05	Adopt or reject engineered defaults	Negative LR and RF engineered-feature results support raw defaults; features remain experimental. Section 7.5.
D06	Full or reduced demographic inputs	Retain full reference and reduced challenger; similar AP is not fairness proof. Section 7.5.2.
D07	Single score or multi-model comparison	Original five-family AP comparison plus a later tuned SVM challenger and nested evaluation. Sections 8.1–8.4.
D08	AP gaps 0.005, 0.010 or 0.015	Record 0.010 three-model panel; exploratory score proximity, not equivalence. Section 8.5.
D09	Different operating thresholds	Record 0.35 after development trade-offs; no verified economic optimum. Section 9.2.
D10	Collapse votes or expose split decisions	Use 3/2/1/0 votes for ACT/REVIEW/MONITOR/HOLD. Section 9.3.
D11	Conformal guarantee or descriptive agreement	No conformal method retained; no coverage guarantee claimed. Sections 8.6 and 11.3.
D12	TRIAGE or ordinary risk ranking	Capacity evidence does not establish reliable capture or ROI improvement. Section 9.6.
D13	Pristine-test claim or historical replay disclosure	Disclose prior exposure; replay the recorded policy without further selection. Sections 9.4 and 13.3.
D14	Replace secondary lens or retain it	Keep Stable High-Risk Customer Profiling; Atlas supports it. Sections 3.2 and 6.5.

12.1.1 Completed Revision Decisions

Record	Evidence	Decision
R01–R03	Dictionary corrected; SMOTENC evaluated and synthetic categories/service relationships checked.	Keep no-balancing reference; retain negative results and synthetic-data limits.
R04–R05	Tuned and nested SVM AP 0.65546; mean training–validation gap 0.00846.	Retain SVM as an evaluated challenger; do not claim absence of overfitting.
R06	RF engineered AP 0.66134; reduced-demographic AP 0.66547.	No engineered default; reduced inputs remain a candidate for further validation.
R07–R09	Interpretation corrections, eight input checks, historical-test disclosure.	Reject common malformed inputs; preserve historical scores and their limitations.
R10–R12	Lecture coverage, parameter guards and explicit execution scope.	Distinguish completed experiments, retained historical outputs and optional methods.

12.2 Changes from the Initial Proposal and Their Reasons

Original proposal	Final implementation	Reason
Risk/stability concept matrix	Explicit three-model vote rule at 0.35	Makes customer states reproducible and auditable.
Illustrative 2% ROC-AUC proximity	Absolute mean-fold AP gap of 0.010	Uses the recorded minority-class ranking metric; exploratory change, not a pre-registered criterion.
Domain-inspired feature hypotheses	Tested but excluded from raw defaults	Engineered sets did not improve the tested LR or RF AP.
Stable High-Risk Customer Profiling	Retained; Atlas added as supporting analysis	Connects descriptive profiles to the original retention lens.
Potential benefit from stability	Risk ranking with agreement as review context	Equal-budget comparisons did not establish reliable improvement.

12.3 Assignment Requirements and Evidence Locations

Required evidence [1, 2]	Report section / GitHub evidence path
Problem-framing canvas	Section 6.1 framing table; Sections 1–4 define lenses, task, stakeholder and output.
Workflow diagram	Section 5 Diagram B and Section 10 handoff table.
Decision log	Section 12.1; original consolidated log and classification_revision_decisions.csv in decision_logs/project/.

Required evidence [1, 2]	Report section / GitHub evidence path
EDA insight log	Section 6.3 and decision_logs/member1/EDA_INSIGHT_LOG.csv.
Preprocessing/feature log	Section 7 and decision_logs/member2/preprocessing_decision_log.csv and decision_logs/member2/feature_registry.csv.
Dataset/data dictionary	Appendix A and dataset/DATA_DICTIONARY.csv; original CSV and saved splits in dataset/.
Model/method comparison	Sections 7–9; MODEL_METHOD_COMPARISON.csv and results/classification_revision/ in model_method_comparison/.
Recommendation and limitations	Sections 9.7–9.8, 11.3 and 14.
Final notebook/code and reproducibility	Section 13.2; final_notebook_code/ including the new-customer demo.
AI-use transparency	Section 13.3.
Video and individual learning journeys	The GitHub link is provided in Section 13.1. The YouTube demonstration is linked in Section 13.1. Each member separately submits a one-page A4 Personal Learning Journey.

13. Reproducibility and Submission Links

13.1 Project Files and Submission Links

Project files (GitHub): https://github.com/Zacky-Ahmed/TRIAGE_Telco_Project

Public dataset reference (Telco Customer Churn):
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Three-minute YouTube demonstration: https://youtu.be/2ilwdmJGN-I

The submission package contains the four required evidence categories: final notebook/code, dataset/data dictionary, decision logs and model/method comparison. The exact CSV is dataset/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv. The repository URL above is the distribution location; this report describes the corrected package, TRIAGE_Telco_Corrected_Submission.zip. The YouTube demonstration is linked above, and each member submits an A4 Personal Learning Journey separately [1, 2].

Folder	**Contents**
final_notebook_code	18 notebooks, shared pipelines, revised SVM/SMOTENC/nested-CV scripts, runners, verification and scoring code.
dataset	Original CSV, data dictionary, saved split/fold membership and clearly labelled synthetic demo inputs.
decision_logs	Original decisions, completed revision decisions and lecture_alignment.csv.
model_method_ comparison	Model comparisons, figures, handoffs, recorded outputs and verification evidence.

13.2 Repository Contents and Execution

Open final_notebook_code/Final_Analysis.ipynb for integrated findings. The member notebooks retain the original workflow; Member 3's 06_classification_revision.ipynb explains the completed SVM, SMOTENC and nested-validation results. Verification used Python 3.13.5 and the pinned requirements. From the extracted project root, run:

python -m pip install -r final_notebook_code/requirements.txt

python final_notebook_code/run_all.py

The runner covers 18 notebooks and supporting scripts. It reconstructs models, replays historical evidence, checks new-customer scoring and runs the classification extension. This revision freshly executed the 14 comparison branches, SVM grid and nested search, synthetic/input audits, corrected Member 1 EDA and summary notebooks. Earlier training and seed outputs were retained where settings were unchanged; this is not a claim that every historical fit was rerun or that a fresh clean-environment reproduction was completed.

The recorded revision passed 67 independent checks, 33 cross-stage checks and 499 release checks. These verify specified inputs, outputs and package integrity; they do not certify real-world accuracy. Source hashes and execution scope are recorded in model_method_comparison/verification/CLASSIFICATION_REVISION_VERIFICATION.json.

To rerun only the completed classification extension after installing the requirements:

python final_notebook_code/run_all.py --stage revision

To run only the new-customer example after installing dependencies, use:

python final_notebook_code/rebuild_frozen_models.py

python final_notebook_code/predict_new_customers.py

To score a different CSV, provide --input customers.csv --output predictions.csv to predict_new_customers.py. Supply the 19 predictor columns and optionally customerID; do not supply Churn. Reconstructing the saved models is a setup step, while scoring an incoming record does not retune or refit models.

13.2.1 Supplementary Figure Reproduction

The included script final_notebook_code/scripts/generate_eda_evaluation_figures.py regenerates descriptive associations and historical ROC/PR and confusion-matrix diagnostics. Its outputs are under model_method_comparison/results/supplementary_descriptive_visuals/. The historical figures replay saved probabilities; they are not new training or independent test evidence.

13.3 AI Assistance and Acknowledgements

AI tools assisted with planning, debugging, code review, integration, report editing and packaging. Members remain responsible for verifying results and accurately reporting their contributions. The IT3091 materials, dataset and research are acknowledged; their authors do not endorse the TRIAGE rule.

14. Final Conclusion

TRIAGE-Telco connects four stages: understanding customer churn, preparing reliable inputs, comparing models and investigating disagreements, and testing what the predictions mean for a retention team. The secondary lens profiles the customers selected by the decision rule; it cannot identify causes of churn or prove which retention offers will work.

Tuned CatBoost led the original development comparison (AP 0.6727) and had historical-test AP 0.6658. Tuned SVM reached nested outer AP 0.6555, while SMOTENC reduced AP in all three tested families. The three-model probability panel remains unchanged. Its four states make agreement visible, but the experiments did not establish better fixed-budget capture than ordinary risk ranking. Even unanimous predictions can be wrong.

We recommend risk-based contact priority and agreement as optional review context, initially in a controlled pilot. Tuning improved the SVM's training–validation gap, but no experiment proves the absence of overfitting. Full-data EDA, historical test exposure, reused original selection folds and missing intervention outcomes remain limitations. Independent future-period evaluation and measured business outcomes are needed before claiming improved retention or financial return.

Appendix A - Data Dictionary

Each row represents a customer. The original dataset contains 7,043 records and 21 columns: customerID, 19 predictors and Churn. The final reference uses the original 19 predictors; demographic-reduction and engineered-feature alternatives were evaluated separately. The machine-readable data dictionary is at dataset/DATA_DICTIONARY.csv.

Field	Business meaning	Values, units and use
customerID	Unique customer identifier	Unique text; tracking only, excluded from predictors.
gender	Customer gender	Female / Male; predictor in full reference.
SeniorCitizen	Senior-citizen indicator (0/1)	0 / 1; binary category stored numerically.
Partner	Whether the customer has a partner	Yes / No; categorical predictor.
Dependents	Whether the customer has dependents	Yes / No; categorical predictor.
tenure	Months with the provider	Nonnegative whole months required for new-customer scoring.
PhoneService	Whether phone service is active	Yes / No; categorical predictor.
MultipleLines	Multiple phone lines status	Yes / No / No phone service.
InternetService	Internet service type	DSL / Fiber optic / No.
OnlineSecurity	Online security add-on status	Yes / No / No internet service.
OnlineBackup	Online backup add-on status	Yes / No / No internet service.
DeviceProtection	Device protection add-on status	Yes / No / No internet service.
TechSupport	Technical support add-on status	Yes / No / No internet service.
StreamingTV	Streaming TV add-on status	Yes / No / No internet service.
StreamingMovies	Streaming movies add-on status	Yes / No / No internet service.
Contract	Customer contract type	Month-to-month / One year / Two year.
PaperlessBilling	Whether paperless billing is used	Yes / No; categorical predictor.
PaymentMethod	Payment method	Electronic check / Mailed check / Bank transfer (automatic) / Credit card (automatic).
MonthlyCharges	Current monthly charge	Numeric; monthly charge in source units.
TotalCharges	Cumulative charges stored as text in raw file	Numeric text converted to number; cumulative charge in source units.
Churn	Whether the customer churned	Yes → 1 / No → 0; target only, never a predictor.

The labels “No internet service” and “No phone service” are meaningful categories. The 11 blank TotalCharges values occur at zero tenure and receive the recorded zero treatment. The scoring entry point checks documented category values, binary SeniorCitizen, whole-month tenure and basic service consistency. Nonblank invalid charges are rejected. The dataset does not establish a currency or every possible business plausibility constraint.

ServiceCount, SupportGap, AutoPay and ChargePerService were tested as extra features but are not required by the final models. Customer demographics and other associations in the report do not establish causal retention actions.

References

[1] Sri Lanka Institute of Information Technology, IT3091 Machine Learning Group Assignment Descriptor, 2026, sections 4–6.

[2] IT3091 Courseweb, Final Submission notice. Guided Data Track requirements include the group report and evidence, a three-minute YouTube demo and one A4 Personal Learning Journey per student.

[3] IBM, Telco Customer Churn example project, https://github.com/IBM/telco-customer-churn-on-icp4d; public Telco Customer Churn dataset reference, https://www.kaggle.com/datasets/blastchar/telco-customer-churn. The coursework CSV is provided at dataset/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv in the linked GitHub repository (Section 13.1).

[4] C. Marx, F. Calmon and B. Ustun, "Predictive Multiplicity in Classification," Proceedings of the 37th International Conference on Machine Learning, PMLR 119, pp. 6765–6774, 2020. https://proceedings.mlr.press/v119/marx20a.html

[5] scikit-learn developers, average_precision_score documentation. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html (accessed 10 October 2026). The repository requirements file pins scikit-learn 1.8.0; live documentation may describe a later release.

[6] Sri Lanka Institute of Information Technology, IT3091 Machine Learning lecture materials, Lectures 1–7 supplied with the project, 2026.